

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027

Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

TherAlign Health, Founding Software Engineer Intern | *Remote* Jul 2026 – Present

- Founding engineer at an early-stage healthtech startup, building full-stack features end to end (Next.js/React front end, Firebase/GCP backend) for a provider-facing **SMART on FHIR** clinical decision support app under fast-paced, ambiguous startup conditions.
- Extending a **Firebase Cloud Functions** backend that builds medication alternatives from **RxNorm** normalization, formulary and prior-authorization logic, and PubMed/guideline evidence, with **Gemini** constrained to synthesis while the backend deterministically controls eligibility and coverage results.

Culture & Morality Lab, UMass Amherst, Lab Assistant | *Amherst, MA* Jul 2026 – Present

- Sole engineer building the lab's text-analysis platform (**Python**, **FastAPI**, sentence-transformers, React), productionizing a published NLP method into a self-serve web tool with full ownership of architecture and technical decisions delegated by the PI.
- Built a YAML model registry, a versioned construct library (**99 constructs from 38 questionnaires**), and a structured data-quality warning system; grew the backend suite to **40 tests** running hermetically in CI.

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clinical clients**, processing **10,000+ monthly triage submissions** via Mayden iaptus REST integrations; contributed to **\$340K+** in revenue.
- Architected a full-stack annotation platform (**React**, Node.js, MongoDB) used daily by the AI team, cutting manual annotation work by **100+ hours/month** and lifting labeling throughput by **60%**.
- Diagnosed and resolved production issues across tenants, monitoring logs during releases and client integrations and shipping targeted fixes for tenant-specific incidents to keep the multi-tenant eTriage service stable.

Cario Growth Services, Backend Developer Intern | *Chennai, India* Nov 2022 – May 2023

- Designed **20+ production RESTful APIs** (Node.js, Fastify, **PostgreSQL**) with input validation and clear contracts for the frontend team; migrated the core codebase from **JavaScript** to **TypeScript** across repositories.

Projects

Cache- and SIMD-Aware Matrix Multiplication (GitHub) C++, AVX/SIMD, OpenMP, Linux perf

- Optimized **C++** matrix multiplication from a naive baseline through cache-aware tiling, manual **AVX/SIMD** vectorization with register-aware micro-kernels, and **OpenMP** multithreading, profiling L1/L3/TLB misses with **Linux perf** and inspecting generated assembly.

Spark ETL Performance Optimization (GitHub) PySpark, Spark UI, SQL, NYC TLC Dataset

- Cut end-to-end Spark ETL runtime **~25%** (17–19 min to 14.3 min) on a **~1.4B-row** NYC TLC dataset via broadcast joins, Adaptive Query Execution, skew-join handling, and shuffle-partition tuning; profiled stage-level timings in the Spark UI to localize bottlenecks.

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a multi-stage agentic web-discovery service (Python, FastAPI) turning open-ended queries into structured entity tables via typed retrieval planning, reranking, and deterministic extraction with LLM fallback and **cell-level provenance**; hardened with **150+ tests**.

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages & Databases: C++, Python, JavaScript, TypeScript, SQL, Java, Bash, PostgreSQL, MongoDB, SQLite

Backend & Web: React, Node.js, Express, FastAPI, REST APIs, Git, GitHub Actions

Systems & Perf: Linux, AVX/SIMD, OpenMP, multithreading, perf / Cachegrind, cache-aware optimization, Spark / PySpark

ML & AI: PyTorch, sentence-transformers, LLM agents, RAG, evaluation

Cloud & DevOps: AWS, GCP, Azure, Docker, CI/CD, MongoDB Atlas